

Recurrent Line Oriented Flight Training (RLOFT) Student Briefing

ATLAS AIR
WORLDWIDE



This presentation is to be used as a tool, to provide a facilitated briefing for our crewmembers.

The goal of this presentation is not to read the slides verbatim.

ATLAS AIR
WORLDWIDE

FTSP Photo Submission Requirement

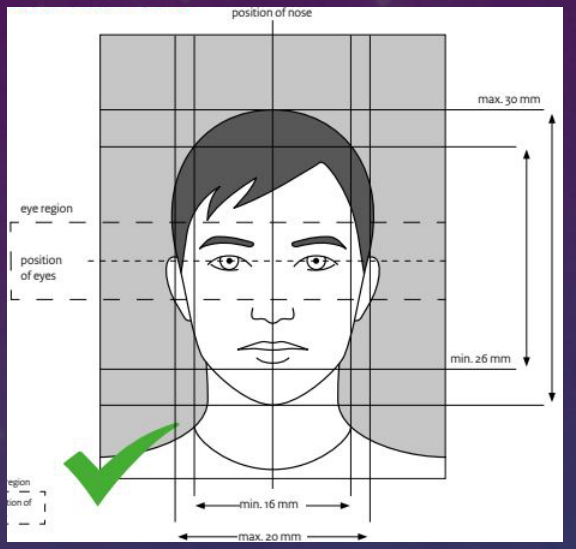
When to Take the Photo?
Instructors: **You must take a photograph of any non-U.S. citizen, non-U.S. national, or lawful permanent resident (Green Card holder)** when they arrive for flight training in a simulator, even if they recently trained with you.

A new photo is required once per curriculum (e.g., Transition, Requalification, and Recurrent are all separate curricula).

Uploading the photograph confirms that you have **exactly matched the individual's identification** to the information shown on their passport.

Instructors: Scan the QR code to access the FTSP Photo Submission Form or tap the link on your iPad.

Atlas Air FTSP Photo Submission



Ensure the photo captures the full face without shadow and that the image is not too dark or blurred.

Focus just on the face; **the image should not include anything below the person's neck.**

FTSP

PRIVACY.FLOWCODE.COM

Photo Submission

SMS data has provided areas where our 747 crews have had issues or knowledge gaps. Reviewing this material provides our crews with an opportunity to evaluate their understanding and application of procedures under different non-normal conditions.

Hot Topic Items

- **Updated OFP**
- **Carriage of Animals and Perishables Review**
- **GPS Jamming/Spoofing**
- **Adverse Weather De/Anti-icing Flap Usage and Before Takeoff Procedure**
- **Updated Fuel Policy**
- **Thrust Reverser Usage**
- **Discontinued/Missed Approach Procedures**
- **Anatomy of a Landing**

Updated OFP

All the computations remain the same as in the old format, but the labels are different.

- **CONT = unplanned contingency fuel which is 10% reserve fuel required by on international flights.**
- **FRES = FAA reserve fuel (45 min domestic, 30 min international)**
- **PCF = dispatcher planned contingency fuel**
- **TOF = takeoff fuel (same as old format MINF). The change here is that MINF value is not rounded up any more so that the math actually works out on the OFP. In this example, the MINF is still 90.7 since the fuel gauge is only visible to a single decimal point. The "rounded value" has always been in "EXTRA" and is now fully visible.**
- **DOW = Dry Operating Weight (includes all cockpit/cabin crew, ACMs and SWA).**
- **SWA = if applicable, will be shown in (parenthesis). This way crews can see the value, but have an indicator that it is already accounted for in the DOW so we don't have to do as much math with the Loadsheet comparison.**

OLD FORMAT

RKSO-PANC

ALT1/2	PAED/		
	FUEL	TIME	DIST
TRIP	**075376**	**07.11**	**3662**
ALT1	003305	00.17	0068
FAR RSV	003997	00.30	
10% RSV	006500	00.44	
ADD			
CF	001500	00.10	
BALLAST			
MINF	**090700**	08.52	....
EXTRA			
TAXI	000900	00.20	
BLOCK	**091600**		
REMF	**015324**	01.55	....

	Weight	Actual	Limit
BOW	181276		
SWA	004154		
PYLD	059600		
ZFW	245030	.....	S251743
RAMP	336630		S395986
TOGW	335730	.....	L361139
LDGW	260354		S285763

NEW FORMAT

RKSO-PANC

ALT1/2	PAED/		
	FUEL	TIME	DIST
TRIP	**075376**	**07.11**	**3662**
CONT	006500	00.44	
ALT1	003305	00.17	0068
FRES	003997	00.30	
PCF	001500	00.10	
MINF	**090678**	08.52	....
EXTRA	000022	00.00	
TOF	090700		
TAXI	000900	00.20	
BLOCK	**091600**		
REMF	**015324**	01.55	....

	Weight	Actual	Limit
DOW	185430		
SWA	(004154)		
PYLD	059600		
ZFW	245030	.....	S251743
RAMP	336630		S395986
TOGW	335730	.....	L361139
LDGW	260354		S285763

Carriage of Animals and Perishables

CARRIAGE OF ANIMALS AND PERISHABLES (FREIGHTER)

Temperature Guidelines

Use the following temperature guidelines in the absence of customer-specified temperatures:

Livestock (cattle, sheep, horses, pigs, etc.)	10-18° C
Domesticated animals (dogs, cats, etc.)	18-21° C
Poultry (chicken, turkeys, ducks, other birds.)	15-21° C
Live plants or trees	13-15° C
Fresh produce (vegetables, flowers, fruit, etc.)	7-10° C
Fresh fish	3-5° C

Set the correct temperature on the TEMP selector for the compartments containing animals and/or perishables.

Livestock and Animals

High levels of heat and moisture can accumulate rapidly both on the ground and in-flight, when carrying large volume of livestock or animals. The following procedures are to be used, to reduce heat and humidity on the ground and in-flight.

Operate all packs continuously. If APU is inoperative, ground conditioned air is required for prolonged ground operations. Every effort should be made to "stay ahead" of temperatures by ventilating and providing cold conditioned air to all compartments where livestock are loaded.

A darkened cargo compartment is preferable, because decreased animal activity reduces metabolic output. One exception, birds will not feed in the dark. Therefore, the cargo compartment lights shall be left on.

Procedures

Documentation

Confirm flight plan trip fuel burn adjustment of: 4 +1%, 6 +.63%.

Arrival At Aircraft

If OAT not excessive:

Nose cargo door ..... Open

Side cargo door ..... Open

Lower lobe cargo doors ..... Open

The Company has incurred significant cost penalties due to the loss of temperature-sensitive cargo.

Care must be taken to correctly follow the procedure and maintain vigilance during flight operations

Carriage of Animals and Perishables

Upon arrival to a transit station or at the cargos' destination, delay turning off the APU and shutting down packs until temperature sensitive cargo has been offloaded

Postflight

Accomplish after offload is complete:

Pack HI FLOW switch ..... Off

4 LOWER LOBE CARGO CONDITIONED AIR FLOW RATE selector ..... OFF

8 LOWER LOBE TEMP selectors (FWD & AFT) ..... OFF

Preflight

ECS synoptic switch ..... Push

APU ..... ON

PACKS (all) ..... On

MAIN DECK TEMP selectors (FWD & AFT) ..... Set

Note: *8The magenta target temperature only shows if at least one of the Forward or Aft Lower Lobe Temperature selectors are in AUTO.*

4 LOWER LOBE CARGO CONDITIONED AIR FLOW RATE selector ..... Set

4 If only aft or forward lower lobe compartment ..... AFT HIGH or FWD HIGH

4 Both lower lobe compartments ..... BOTH LOW

LOWER LOBE TEMP selectors (FWD & AFT) ..... Set

If carrying animals

Coordinate that all cargo doors remain open until just prior to pushback.

Before Start

PACKS 2 & 3 ..... OFF

Before Taxi

PACKS (all) ..... On

Pack HI FLOW switch ..... ON

4 AFT CARGO HEAT switch ..... As required

Cruise Altitude

If main deck temperature is warmer than lower lobe:

4 LOWER LOBE CARGO CONDITIONED AIR FLOW RATE selector ..... OFF

8 LOWER LOBE TEMP selectors (FWD & AFT) ..... OFF

When main deck is equal to lower lobe, return selectors to their previous position(s).

10 Minutes Prior To T/D

If carrying animals

Coordinate an early descent with a reduced rate with ATC. Early descent allows thrust setting above idle for increased pack output and improved temperature control during descent.

Descent ..... Initiate (with ATC clearance)

Postflight

Accomplish after offload is complete:

Pack HI FLOW switch ..... Off

4 LOWER LOBE CARGO CONDITIONED AIR FLOW RATE selector ..... OFF

8 LOWER LOBE TEMP selectors (FWD & AFT) ..... OFF

GPS Jamming/Spoofing

Have you had any issues?
What have been your indications?

>TRANSPONDER L, R

Condition: One of these occurs:
- The transponder is failed
- ADS-B Out function is inoperative

The ADS-B Out function can be inoperative due to loss of valid GPS data. The transponder is operative, but ADS-B Out is inoperative, when all of these are true:
- A TCAS mode is selected
- TCAS OFF and TCAS SYSTEM advisory messages are blank

GPS signal reception can improve as the airplane location changes. When valid GPS position data is available, ADS-B operation is restored and the message blanks.

■ ■ ■ ■

Unannunciated Indications

GPS Data Unreliable

Condition: Erroneous or invalid GPS data is suspected.

Caution! GPS interference can alter the airplane flight path without obvious flight deck indications or effects.

1 Choose one:

◆On the **ground**:

Use Supplementary Procedures or operator-originated standard operating procedures for intentional operations in areas of known GPS interference .

■ ■ ■ ■

◆In **flight**:

►►**Go to step 2**

2 One or more of the following can indicate unreliable GPS data:

PULL UP alert that occurs with sufficient, immediately verifiable terrain clearance

Change in terrain data on the ND that does not match aeronautical charts

Slow increase in ANP

FMC position on the ND changes to DD or IRS

Excessive lateral deviation when intercepting the localizer or final approach course

Inaccurate fuel predictions

Unable to log on to ATC Data link with inaccurate UTC TIME/DATE

▼ Continued on next page ▼

▼GPS Data Unreliable continued▼

Inaccurate UTC TIME/DATE and FMC-generated ETA or RTA

The following alerts can occur, and must be considered valid:

GPS

GPS LEFT, RIGHT

GND PROX SYS

TERR POS

TRANSPONDER L, R

UNABLE RNP

3 Choose one:

◆Indications of unreliable GPS data **occur**:

►►**Go to step 4**

◆Indications of unreliable GPS data do **not** occur:

■ ■ ■ ■

4 GPS FMC position updating and GPWS terrain look-ahead functions can be temporarily inhibited.

5 Choose one:

◆GPS FMC position updating inhibit is **needed**:

►►**Go to step 6**

◆GPS FMC position updating inhibit is **not** needed:

►►**Go to step 9**

▼ Continued on next page ▼

▼GPS Data Unreliable continued▼

6 NAV RAD key . . . . . .Push

Shows the NAV RADIO 1/2 page.

7 NEXT PAGE key . . . . . .Select

Shows the NAV OPTIONS 2/2 page.

8 GPS UPDATE> . . . . . .Select

Verify OFF shows in large font.

Select DME/VOR/LOC UPDATE to ON, as needed.

Monitor GPS FMC position updating for recovery.

Note: GPS data recovery can be indicated by verifying both of the following after selecting POS REF page 3:

- GPS LAT/LON shows for GPS L or GPS R

Selecting BRG/DST on POS REF page 3

- Distance shown for GPS L or GPS R is less than current ANP, based on inertial or radio nav data on ND

If GPS data appears to recover, turn GPS UPDATE> to ON at Captain's discretion.

▼ Continued on next page ▼

Supplemental Procedure 11.13 Review

GPS Jamming/Spoofing

NAVAID INHIBIT

Note: *Inhibit GPS updates for approach operations that are not based on WGS-84, unless other appropriate procedures are used..*

NAV RAD key ..... Push
Shows the NAV RADIO 1/2 page.

NEXT PAGE key ..... Select
Shows the NAV OPTIONS 2/2 page.

To inhibit GPS updates:

GPS updating can be inhibited at pilot discretion if it is not needed or GPS data is unreliable. It is not recommended to disable GPS updating for the entire flight.

Note: *GPS updating is needed for some mandatory navigation specifications.*

Note: *Inhibiting GPS updating does not inhibit the GPWS look-ahead terrain function. The GPWS look-ahead terrain function can be inhibited by selecting TERR OVRD to OVRD on the Ground Proximity Panel.*

GPS UPDATE> ..... Select
Verify OFF shows in large font.

Note: *If GPS UPDATE> is selected OFF, it is recommended that DME UPDATE, VOR UPDATE, and LOC UPDATE be selected ON.*

Note: *Airplanes with DME updates defaulted to OFF should select DME UPDATE to ON. Selecting VOR UPDATE to ON, in addition to selecting DME updates to ON, will provide VOR/DME radio updates.*

To inhibit DME updates (DME/DME):

Note: *VOR/DME updates will also be inhibited.*

DME UPDATE> ..... Select
Verify OFF shows in large font.
Verify ALL shows in 1L and 1R.

To inhibit VOR updates (VOR/DME):

VOR UPDATE> ..... Select
Verify OFF shows in large font.
Verify ALL shows in 2L and 2R.

To inhibit LOC updates:

LOC UPDATE> ..... Select
Verify OFF shows in large font.

Supplementary Procedures -
Flight Management, Navigation

To inhibit a navaid (for one or two navaids):

Enter the navaid identifier in the scratchpad.

NAVAID INHIBIT (1L or 1R) ..... Enter

To inhibit a VOR (for one or two VORs):

Enter the VOR identifier in the scratchpad.

VOR ONLY INHIBIT (2L or 2R) ..... Enter

GPS Jamming/Spoofing

How do we correct this in flight?

When a flight crew encounters GPS signal interference, report the event to ATC and Dispatch as soon as practical.

- **Verify POS initially by selecting POS on EFIS Panel**
- **Crosscheck CDU POS REF pages**
- **Select the most accurate if available**
- **Inhibit GPS if necessary**

The GPWS Look-Ahead Terrain, and the Automatic Dependent Surveillance (ADS)/Air Traffic Services (ATS) systems utilize GPS data, and GPS interference can affect these systems.

Time permitting, reference:
-747 Fleet Captains Notes 11/2023
-Boeing FOTB 747 23-80 R1 on Comply, pages 7-14 for indication reference

When De/Anti-icing is Required (Off Parking Stand)

If there is snow or ice accumulation on the wing, consider delaying the flight control check until after deicing/anti-icing is accomplished.

If taxi route is through ice, snow, slush, or standing water in low temperatures or if precipitation is falling with temperatures below freezing, taxi out with the flaps up. Taxiing with the flaps extended subjects the flaps and flaps drives to contamination. Leading edge flaps are also susceptible to slush accumulations.

Call “FLAPS ___” as needed [C].

Flap lever [F]..... Set flaps, as needed

SP.16.7

After De/Anti-icing is Completed (Off Parking Stand)

After de/anti-icing is completed:

Calculate holdover time if applicable.

WI-FI switch (Gogo Internet) ***(Passenger)*** [F] ..... ON

APU [F] .....As needed

APU bleed air switch (APU running) [F]..... ON

Choose one:

If engines **running:**

Wait approximately one minute after deicing is completed to turn pack selectors on to ensure all deicing fluid has been cleared from the engines.

PACKS control selectors/switches [F]..... 4 NORM / 8 ON

Flight controls [C] .....Check

Do the Before Taxi Checklist.

SP.16.10

Before Takeoff Procedure (De/Anti-icing Off Parking Stand)

Before Takeoff Procedure

Do the normal Before Takeoff Procedure with the following modification:

Call “FLAPS ___” as needed for takeoff. [PF]

Flap lever [PM] .....Set takeoff flaps, as needed

Extend the flaps to the takeoff setting at this time if they have been held due to slush, standing water, or icing conditions, or because of exterior de/anti-icing.

Do the Before Takeoff Checklist.

SP.16.11

FMC Reserves (Planned Diversion Fuel)

- FMC Reserves (Planned Diversion Fuel) is defined as the fuel required to fly to the alternate and thereafter for 45 minutes holding at 1,500 feet AGL. FMC Reserve on the OFP provides the minimum diversion fuel from the destination airport.
- By programming the FMC Reserve fuel, the FMC will generate a scratchpad message when the estimated fuel at the final destination airport falls below the FMC Reserve value.

Note: The Insufficient Fuel scratchpad message is based on predicted FMC fuel burn and programmed routing from your present position to the destination airport.

FMC Reserves (Planned Diversion Fuel)

FMC Reserves (Planned Diversion Fuel) is defined as the fuel required to fly from the destination airport to the alternate and thereafter for 45 minutes holding at 1,500 feet AGL. FMC Reserve on the OFP provides the minimum diversion fuel from the destination airport (including ballast fuel if applicable).

By programming the FMC Reserve fuel, the FMC will generate a scratch pad message when estimated fuel at the final destination airport falls below the FMC Reserve value. Refer to *In-flight Fuel Considerations, Contingency Planning and Use of Alternate or Planned Diversion Fuel* on page 10.1.37.

It is important to note that reaching FMC Reserves (Planned Diversion Fuel) does not require a "MINIMUM FUEL" declaration to ATC. Refer to *ATC Minimum Fuel Advisory* on page 10.1.39.

NOTE

This fuel figure is different than the RESERVES fuel number included in a Redispatch Message (RDM). See *Required Fuel* on page 7.1.31.

MINIMUM vs MAYDAY Fuel Declaration FOM Ver 27 10.1.41

Table 10.1.2. Minimum and MAYDAY Fuel Declaration

MINIMUM AND MAYDAY FUEL DECLARATION	
Step 1 Request delay information when required	Seek information from ATC concerning any expected delays to assist in determining your committed landing aerodrome.
Step 2 Declare MINIMUM FUEL when committed to land at a specific aerodrome and any change in the existing clearance may result in a landing with less than planned Final Reserve Fuel	Declare "Minimum Fuel" which should represent the last lines of defense in a multilayered strategy designed to ensure the protection of **Final Reserve Fuel** and safe flight completion. Requires the PIC should declare "MINIMUM FUEL" when, based on the current ATC clearance, the anticipated amount of fuel remaining upon landing at the aerodrome to which the aeroplane is committed is approaching the planned **Final Reserve Fuel** quantity. This declaration is intended to convey to the applicable air traffic controller that so long as the current clearance is not modified, the flight should be able to proceed as cleared without compromising the PIC's responsibility to protect **Final Reserve Fuel**.
Step 3 Declare a fuel emergency when the calculated fuel on landing at the nearest suitable aerodrome, where a safe landing can be made, will be less than the planned **Final Reserve Fuel**	Declare Fuel Emergency using "**MAYDAY, MAYDAY, MAYDAY, FUEL**". The last in a series of procedural steps to ensure the safe completion of a flight is the declaration of an emergency. Requires the PIC to declare a situation of emergency by broadcasting MAYDAY, MAYDAY, MAYDAY, FUEL when the calculated usable fuel to be available upon landing at the nearest suitable aerodrome where a safe landing can be made will be less than the planned **Final Reserve Fuel**. This declaration provides the clearest and most urgent expression of an emergency situation brought about by insufficient usable fuel remaining to protect the planned **Final Reserve Fuel**. It communicates that immediate action must be taken by the PIC and the air traffic control authority to ensure that the airplane can land as soon as possible. The "MAYDAY" declaration is used when all opportunities to protect **Final Reserve Fuel** have been exploited and in the judgment of the PIC, the flight will now land with less than **Final Reserve Fuel** remaining in the tanks. The word FUEL is used as part of the declaration simply to convey the nature of the emergency to ATC. It is also important to note an emergency declaration not only opens all options for pilots (e.g. available closed runways, military fields, etc.) but it also allows ATC added flexibility in handling an airplane.

ATC Minimum Fuel Advisory

- The term “MINIMUM FUEL" is used to describe a situation in which an aircraft's fuel supply has reached a state where little or no delay can be accepted. On initial contact with ATC, the term "minimum fuel" should be used after the call sign: (handling but "Minneapolis Approach, Giant 917 Heavy, minimum fuel")
- FOM 10.1.38: The “MINIMUM FUEL” declaration informs ATC that all planned airport options have been reduced to a specific airport of intended landing. This is not handling but but an indication that an emergency situation is possible, should any additional delay occur
- A “MINIMUM FUEL” advisory will not necessarily give crews priority handling but notifies ATC of the current fuel status. A declaration of “MINIMUM FUEL” is not meant to be used to avoid a diversion, it is used after other options have been exhausted and the flight is committed to landing at a specific airport.

ATC Minimum Fuel Advisory

ICAO Annex 6 4.3. 7.2. 2

The pilot-in-command shall advise ATC of a minimum fuel state by declaring MINIMUM FUEL when, having committed to land at a specific aerodrome, the pilot calculates that any change to the existing clearance to that aerodrome may result in landing with less than planned Final Reserve Fuel. ICAO phraseology guidance is "[CALL SIGN], MINIMUM FUEL". Do not expect priority this is only intended to be an advisory to ATC.

MAYDAY Fuel Declaration

Emergency Fuel

ICAO Annex 6 4.3. 7.2.3

The pilot-in-command shall declare a situation of fuel emergency by broadcasting MAYDAY, when the calculated usable fuel predicted to be available upon landing at the nearest aerodrome where a safe landing can be made is less than the planned Final Reserve Fuel. ICAO phraseology guidance is "[CALL SIGN], MAYDAY, MAYDAY, MAYDAY, FUEL".

There should be no delay in declaring a distress situation (mayday) due to low fuel. Pilots or Dispatchers should not hesitate to make this declaration and to receive priority handling if, in their opinion, delays beyond those causing the minimum fuel condition are likely.

If emergency fuel is reached, the pilot should:

- Declare a distress situation (mayday) with ATC
- State the usable fuel in time remaining, and souls on board
- Request priority handling from ATC
- Proceed directly to the airport
- If time permits, advise the flight Dispatcher of the emergency condition

When an emergency fuel condition exists, the Dispatcher should:

- Provide any assistance to the flight crew (including the declaration of a distress situation (mayday) with ATC if appropriate)
- Ensure all appropriate ATC facilities are advised of the emergency
- Ensure that the airport of intended landing is advised of the emergency
- Ensure that all appropriate emergency procedures and notifications are initiated

Pop Quiz
Hot Shot!

You are planned ANC to ICN with NRT as an Alternate.

If you are holding outside of ICN and an "Insufficient Fuel" scratchpad message appears.

Q: Is it acceptable to declare "Minimum Fuel" to continue toward ICN?

Pop Quiz
Hot Shot!

You are planned ANC to ICN with NRT as an Alternate.

If you are holding outside of ICN and an "Insufficient Fuel" scratchpad message appears.

Q. Is it acceptable to declare "Minimum Fuel" to continue toward ICN?

A. NO- Declaring Minimum Fuel is not legal to use if you still have a valid alternate and sufficient fuel to proceed to that alternate.

Pop Quiz
Hot Shot!

Same flight, same holding point outside of ICN.

Weather, traffic and runway configuration all favor the ability to drop your alternate with concurrence from Dispatch to facilitate a longer hold.

Q. How would you calculate your new FMC Reserve or verify a Dispatch created value is accurate?

Pop Quiz
Hot Shot!

	FUEL	TIME
TRIP	**112674**	**09.49**
CONT	002942	00.18
ALT1	(004016	00.23
ALT2	006952	00.42
FRES	004742	00.30

Same flight, same holding point outside of ICN. Weather, traffic and runway configuration all favor the ability to drop your alternate with concurrence from Dispatch to facilitate a longer hold.

Q. How would you calculate your new FMC Reserve or cross check a Dispatch created value is accurate?

A. Take your FRES fuel divided by time shown and multiply that by 45 min. This is your new FMC Reserve.

Example: 4742 ÷ 30 = 158.06
158.06 x 45 = 7113

New FMC Reserve for ICN with no alternate should be approximately 7.2 (Yes, there are other ways to "math" this problem…)

Pop Quiz
Hot Shot!

Same flight, same holding point outside of ICN. Weather, traffic and runway configuration all favor the ability to drop your alternate with concurrence from Dispatch to facilitate a longer hold and a new FMC Reserve value has been entered in the FMC.

Q. If an FMC scratchpad message “Insufficient Fuel” appears, is this the time to declare “Minimum Fuel”?

Pop Quiz
Hot Shot!

Same flight, same holding point outside of ICN. Weather, traffic and runway configuration all favor the ability to drop your alternate with concurrence from Dispatch to facilitate a longer hold and a new FMC Reserve value has been entered into the FMC.

Q. If an FMC scratchpad message “Insufficient Fuel” appears, or just prior to it appearing, is this the time to declare “Minimum Fuel”?

A. Yes: When the aircraft approaches a fuel level that is less than the amount needed to proceed to the committed destination (with no alternate) and hold for 45 minutes at 1500’ AGL, “Minimum Fuel” may be declared.

MAYDAY Declaration

- MAYDAY FUEL is used when all opportunities to protect reserve fuel have been exploited and in the judgement of the PIC, the flight is predicted to land with 30 minutes or less fuel remaining.
- The correct phraseology is "MAYDAY, MAYDAY, MAYDAY, FUEL". This declaration provides the clearest and most urgent expression of an emergency situation brought about by insufficient usable fuel remaining.
- There should be no delay in declaring a MAYDAY FUEL due to low fuel. Pilots or Dispatchers should not hesitate to make this declaration and to receive priority handling if, in their opinion, delays beyond those causing the minimum fuel condition are likely.

FOM Chapter 10 Irregular Operations: Emergency Fuel

Let's Keep it Standard!!

What is Atlas Air's landing performance baseline?

-Minimum use of flaps

-Minimum use of available autobrake selection

- Idle reverse thrust unless THR REV Credit is selected and needed for brake temperature consideration

FCOM, Vol 1; NP.45

Landing Performance Assessment

The ability of the aircraft to land and stop on a particular runway, using current ambient conditions and aircraft status, shall be determined prior to landing.

A landing distance assessment shall be accomplished, prior to each landing, using available performance data to ensure the aircraft landing weight, stopping distance and brake use or setting are appropriate for ambient conditions and aircraft status for the actual runway of landing.

Flaps 25 is preferred due to noise and maintenance. Flaps 30 may offer operational advantages in some situations. Flaps 30 is the preferred CAT II/III landing flap setting.

Note: *When more than IDLE reverse thrust is used and stopping is assured, reducing reverse thrust to IDLE by 80 knots provides FOD mitigation in addition to significant engine wear improvement.*

General guidance for autobrakes setting:

For normal landing conditions, autobrake 2 used with idle reverse thrust optimizes brake wear. When selecting autobrakes, consider landing distance, runway length and condition, exit plan, actual and expected brake temperatures, and weather.

With the exception of operational considerations (i.e., quick turns, environmental issues), use of the autobrake system is recommended for all landings.

Landing Energy Management

Core Principle

Landing Energy Management is an ADM process – not a checklist. FDP provides cues, not directives. The pilot decides how to manage each landing using available tools.

What Drives Brake Energy

- Small speed changes = large brake-energy changes.
- Mass contributes linearly; speed contributes exponentially.
- Most brake energy is absorbed early in rollout when groundspeed is highest.

FDP – Use as a Decision Tool

- Enter expected weight, winds, and runway state.
- If conditions are uncertain, bias toward a worst-credible scenario.
- Use FDP to compare configurations (flaps, reverse, autobrake).
- Re-run FDP when conditions meaningfully change.
- FDP predicts brake temps and stopping distance – these are cues, not limits.

Aeronautical Decision Making (ADM)

FDP Sequence (typical)

Flaps → Reverse Thrust → Autobrake → Runway

- Flaps 25 = baseline; Flaps 30 is often the first and most economical mitigation.
- Reverse thrust is most effective at high speed.
- Select the lowest autobrake setting that meets runway and condition needs.
- Execute the rollout you planned; avoid early turnoffs.

747-Specific Note

- Reverse thrust on the 747 carries a unique FOD risk: engines 2/3 can lift debris into engines 1/4.
- Crews transitioning from twins must understand the geometry-driven FOD hazard unique to the 747.
- Reverse remains a pilot tool when conditions warrant it.
- Crews apply reverse thrust based on ADM and FDP results – not habit.

Lets Keep it Standard!!

747 Fleet Captain Bulletin

Effective: 06 FEB 2025 **Expires: 31 DEC 2025**

Recent Fleet Experience

In the past year, Atlas has had 7 premature engine removals averaging 2,140 cycles of age due to known or suspected runway chemical ingestion. When compared to the expected on wing performance life of this engine (3200 cycles), this equates to approximately $12.5m annually. We currently have some engines flying on watch with the same performance trend characteristics associated with runway chemical ingestion, with 2 engines as low as 249 cycles and 717 cycles since their last scheduled shop visit.







Recommendations

GE recommends the following in order to reduce the probability of runway chemical ingestion in addition to reducing peak temperatures during climb:

1. Deploy reversers normally to idle reverse (normal operations)
2. Modulate reverse thrust up to maximum reverse thrust as necessary for conditions
3. If higher reverse thrust is used, recommend reducing reverse thrust by 80 knots to be at reverse idle by 60 knots to minimize FOD ingestion.
4. Use the climb thrust selected by the FMC unless a higher selection is needed for obstacles, altitude restrictions, ATC, etc. Return to the FMC selected climb thrust when the higher selection is no longer needed.

Mitigation Strategies

When performance calculations allow, crews should consider limiting reverse thrust to idle and using full length landings. When appropriate, minimize thrust reversers at high power settings below 80 knots (idle reverse by 60 knots). This language has been incorporated in FCOM Rev 17

Use the climb thrust selected by the FMC unless a higher selection is needed for obstacles, altitude restrictions, ATC, etc. Return to the FMC selected thrust when the higher selection is no longer needed. This language has been incorporated in FCOM Rev 17

Stowing Thrust Reversers (FCOM NP.55.1)

Do not stow thrust reversers until the engines have stabilized at reverse idle N1.

Stowing thrust reversers prior to reverse idle will result in a significant amount of forward thrust when the thrust reversers close. Depending on the N1 when the thrust reversers are closed, forward thrust could be as high as approx. 55% N1 (400) vs. stowing at idle approx. 35% N1 (400). This is particularly important on slippery or contaminated runways.

Engine data has shown significant damage above normal levels due to overuse of reverse thrust. SOP has changed to start TR movement to idle at 80 kts (or any higher speed) once a safe stop is assured.

Warning !	***After reverse thrust is initiated, a full stop landing must be made. If an engine stays in reverse, safe flight is not possible.***
Warning !	***If reverse thrust credit is needed to meet landing performance, full reverse thrust should be used until stopping is assured.***
Caution !	*Use of high levels of reverse thrust below 80 knots may damage engine(s) on contaminated runways (ice, slush, or FOD) or when engines extend beyond prepared runway surface.*
Without delay, move the reverse thrust levers to the interlocks and hold light pressure until the interlocks release. Apply reverse thrust as needed.	Verify that the forward thrust levers are closed. When all REV indications are green, call "Reversers normal." If there is no REV indication(s) or the indication(s) stays amber, call "No reverser(s) engine number ___" or "No reversers".
If stopping is assured, by 80 knots, start movement of the reverse thrust levers to be at the reverse idle detent by 60 knots.	Call "60 knots"
After the engines are at reverse idle, move the reverse thrust levers full down.	

Stabilized Approach Gate

All approaches have a stabilized approach gate of 1000 ft AGL.

- The gate is a bottom line; there is no wiggle room
- The PF shall execute an immediate go-around for any approach that is not stabilized by 1000 ft AGL, becomes unstable below 1000 ft AGL or the PM announces “GO-AROUND”

FOM 10.1.16 Recommendation

- Crossing the FAF on flight path/configured/on target speed will assist in achieving stable criteria by 1000 ft AGL

Stable/Unstable Approach Callouts

- These callouts challenge us to declare that we are stable:
  - An assessment of the aircrafts state must be made prior to calling “STABLE”
  - Automatic responses (e.g. no aircraft assessment – On flight path, Configured for landing, On speed) defeats the safety protocol

Approach and Landing

- High descent rates (greater the 1000 fpm) can be expected with a high Vref or high terrain (e.g. high density altitude/high ground speed):
  - Expected descent rates greater than 1000 fpm should be briefed. If expected descent rates are briefed, a go around call may not be required for “SINK RATE” (FOM 10.1.16)
- Maintain target speed until approaching the flare. If airspeed is excessive when approaching the flare – go-around
- If airspeed is below VREF crossing the runway threshold – go-around

Missed Approach Duties

GO-AROUND PROCEDURE

Pilot Flying	Pilot Monitoring
Push TO/GA switch.	
Verify that the thrust increases.	
Warning ! ***TO/GA function is inhibited close to the ground and after touchdown. If the thrust levers do not respond when TO/GA is pushed, or if the go-around is initiated after touchdown, advance the thrust levers manually to go-around thrust.***	
Call "Flaps 20" or "Flaps __" as needed.	Set the flap lever as directed.
Verify rotation to go-around attitude.	
	Verify that the thrust is sufficient for the go-around or adjust as needed. Call "Thrust set".
Verify a positive rate of climb on the altimeter and call "Gear Up".	Verify a positive rate of climb on the altimeter and call "Positive Rate". Set the landing gear lever to UP.
Above 400 feet RA, Select or call for a roll mode. (8 Verify or select)	Verify that the missed approach altitude is set.
Verify missed approach route being tracked.	
At 1,000 feet AGL[1], select or call VNAV or FLCH. If FLCH is selected, set speed to the maneuvering speed for the planned flap setting. If VNAV is selected: • select speed intervention as needed • set speed to the maneuvering speed for the planned flap setting [1] Above 400 feet RA if needed.	Select commanded pitch mode
Call "Flaps ___" according to the flap retraction schedule.	Position Flap lever as directed.

New PM call - "Thrust Set"

Select and Verify

Above 400' - Lateral Mode

At 1000' - Vertical Mode

As Needed – Speed Set

Verify Correct Altitude is Set

It should be noted - to prevent an overshoot, a vertical mode may be selected as early as 400 feet if the initial MAA is below or close to 1000 feet.

Discontinued Approach

There have been instances of crews using TOGA on a discontinued approach and “overshooting” the MAA due to thrust and climb rate. This can also occur at minimums when the MAA is in proximity of 1000 feet.

It is important to note - the AFDS always performs a “1G” maneuver for passenger and crew comfort, therefore when near the MAA and TOGA is selected, the thrust and climb rate are such that the aircraft is unable to level off at the MAA.

This is also why VNAV may be selected early (at or above 400 feet) when a go-around is initiated at minimums and the MAA is close to 1000 feet AGL. Selecting VNAV (at 1000 feet per SOP) when near the MAA will result in an “overshoot” due to the “1G” maneuver limitation.

Discontinued Approach – Descent May Be Required

- A descent may be required, if you are above the MAA or ATC assigned altitude.
  - The MCP altitude must be set.
  - V/S could be an appropriate mode (after TOGA or A/P and F/D OFF).
- Or:
  - Provided no vectors are immediately given:
    - Set the MCP altitude.
    - A Non-ILS approach can be continued and level off at the MCP altitude.
    - An ILS cannot be continued and leveled off at the MCP altitude.

Discontinued Approach

- To alleviate TOGA thrust application, the following may be used:
  - Disconnect A/P and A/T
  - Manually fly the aircraft as necessary to adhere to the clearance
  - Both F/D's OFF then ON (If on an ILS) – FMA result - HDG HLD or ATT and V/S
  - Autopilot – as desired

How you perform a Discontinued Approach is up to you

Mismanaged go-arounds

If an overshoot of the MAA is inevitable because the go-around is mismanaged:

- Get back to basics AND FLY THE PLANE.
- Disconnect the A/P, do not let the A/P FLY YOU.
- Manually fly the aircraft to adhere to clearances.
- Once the aircraft is conforming to the clearance, automate as appropriate

Anatomy of a Landing

Safety and FCR data have shown the 5 top contributing factors to having a hard or unstable landing are:

1. **Seat Position**
2. **Height of Autopilot Disengagement**
3. **Visual Transition**
4. **Ground Effect awareness and recognition**
5. **50’ FCTM guidance to touch down**

Anatomy of a Landing Discussion

1. Seat Position FCTM 2.4, FCTM 2.8

What is the Optimal Eye Position?

What must the crewmember be able to view from the seated position? Taxi, Takeoff, and Landing

The pilot's seat should be adjusted for optimum eye position. The rudder pedals should be adjusted so that it is possible to apply maximum braking with full rudder deflection.

During taxiing, the pilot's heels should be on the floor, sliding the feet up on the rudder pedals only when required to apply brakes to slow the taxi speed, or when maneuvering in close quarters on the parking ramp.

2. Height of A/P Disengagement FCOM L.10.7, FCTM 6.8

When is it recommended to disengage the autopilot?

What threats might be identified with low-level autopilot disengagement?

Note: When a manual landing is planned from an approach with the autopilot engaged, the transition to manual flight should be planned early enough to allow the pilot time to establish airplane control before beginning the flare. The PF should consider disengaging the autopilot and disconnecting the autothrottle 1 to 2 nm before the threshold, or approximately 300 to 600 feet above field elevation.

3. Visual Transition FCTM 6.5 + AIM 91.129

Does an instrument approach transition to a visual segment?

How do you transition to the visual segment?

When do you transition to the visual segment?

§ 91.129

(e) ***Minimum altitudes when operating to an airport in Class D airspace.***

(1) Unless required by the applicable distance-from-cloud criteria, each pilot operating a large or turbine-powered airplane must enter the traffic pattern at an altitude of at least 1,500 feet above the elevation of the airport and maintain at least 1,500 feet until further descent is required for a safe landing.

(2) Each pilot operating a large or turbine-powered airplane approaching to land on a runway served by an instrument approach procedure with vertical guidance, if the airplane is so equipped, must:

(i) Operate that airplane at an altitude at or above the glide path between the published final approach fix and the decision altitude (DA), or decision height (DH), as applicable; or

(ii) If compliance with the applicable distance-from-cloud criteria requires glide path interception closer in, operate that airplane at or above the glide path, between the point of interception of glide path and the DA or the DH.

(3) Each pilot operating an airplane approaching to land on a runway served by a visual approach slope indicator must maintain an altitude at or above the glide path until a lower altitude is necessary for a safe landing.

(4) Paragraphs (e)(2) and (e)(3) of this section do not prohibit normal bracketing maneuvers above or below the glide path that are conducted for the purpose of remaining on the glide path.

Anatomy of a Landing Discussion

4. Ground Effect FCTM 6.8 or other aviation aerodynamic references

At what height above ground do we encounter ground effect?

What are the identifiable threats of ground effect if not anticipated?

How might ground effect change a crewmember's control inputs?

Could ground effect interrupt the normal mechanics involved in landing?

5. 50' Radio Altimeter Callout FCTM 6.8

What are the steps the crewmember must have completed for each radio altimeter call starting a 50'?

Avoid rapid control column movements during the flare. If the flare is too abrupt and thrust is excessive near touchdown, the airplane tends to float in ground effect. Do not allow the airplane to float or attempt to hold it off. Fly the airplane onto the runway at the desired touchdown point and at the desired airspeed.

Note: Do not trim during the flare. Trimming in the flare increases the possibility of a tail strike.

Prolonged flare increases airplane pitch attitude 2° to 3°. When prolonged flare is coupled with a misjudged height above the runway, a tail strike is possible. Do not prolong the flare in an attempt to achieve a perfectly smooth touchdown. A smooth touchdown is not the criterion for a safe landing.

When the threshold passes out of sight under the airplane nose shift the visual sighting point to the far end of the runway. Shifting the visual sighting point assists in controlling the pitch attitude during the flare. Maintaining a constant airspeed and descent rate assists in determining the flare point. Initiate the flare when the main gear is approximately 30 feet above the runway by increasing pitch attitude approximately 2° - 3°. This slows the rate of descent.

After the flare is initiated, smoothly retard the thrust levers to idle, and make small pitch attitude adjustments to maintain the desired descent rate to the runway. A smooth thrust reduction to idle also assists in controlling the natural nose-down pitch change associated with thrust reduction. Hold sufficient back pressure on the control column to keep the pitch attitude constant. A touchdown attitude as depicted in the figure below is normal with an airspeed of approximately VREF. Ideally, main gear touchdown should occur simultaneously with thrust levers reaching idle.



SPOT
Special Purpose Operational Training

ATLAS AIR
WORLDWIDE

Special Purpose Operational Training (SPOT)

SPOT is a simulator training session designed to address specific training objectives. Training objectives are based on technical and CRM requirements, and include specific items to be critiqued and debriefed on both technical and CRM performance. SPOT may consist of full or partial flight segments depending on the training objectives for the flight.

At the end of today's RLOFT we will utilize SPOT training to include scenario-based and maneuver-based stall prevention training while completing 3 takeoffs and landings.

Special Purpose Operational Training (SPOT)

Training Exercises to be completed

- Takeoff Departure Stall (390Kg)
- Clean Configuration Stall
- Landing Configuration Stall
- JFK VOR Runway 31L
- Visual Traffic Pattern
- Crosswind Landing (25+ kts)
- Balked Landing
- Landing Currency

SPOT Review – Approach to Stall Recovery

APPROACH TO STALL RECOVERY

All recoveries from approach to stall should be done as if an actual stall has occurred.

Immediately do the following at the first indication of stall (buffet or stick shaker).

Note: *Do not use flight director commands during the recovery.*

Pilot Flying	Pilot Monitoring
Initiate the recovery: • Hold the control column firmly. • Disengage autopilot and disconnect autothrottle. • Smoothly apply nose down elevator to reduce the angle of attack until buffet or stick shaker stops. Nose down stabilizer trim may be needed *	• Monitor altitude and airspeed. • Verify all required actions have been done and call out any omissions. • Call out any trend toward terrain contact
Continue the recovery: • Roll in the shortest direction to wings level if needed.** • Advance thrust levers as needed. • Retract the speedbrakes. • Do not change gear or flap configuration, except • During liftoff, if flaps are up, call "Flaps 1"	• Monitor altitude and airspeed. • Verify all required actions have been done and call out any omissions. • Call out any trend toward terrain contact. • Set the FLAP lever as directed
Complete the recovery: • Check airspeed and adjust thrust as needed. • Establish pitch attitude. • Return to the desired flight path. • Re-engage the autopilot and activate autothrottle, if desired	• Monitor altitude and airspeed. • Verify all required actions have been done and call out any omissions. • Call out any trend toward terrain contact

Warning ! **** If the control column does not provide the needed response, stabilizer trim may be necessary. Excessive use of pitch trim may aggravate the condition, or may result in loss of control or in high structural loads.***

Warning ! ***** Excessive use of pitch trim or rudder may aggravate the condition, or may result in loss of control or in high structural loads.***

SPOT Review - Stall Training Recovery Goals

- Recognizing an approach to stall
- Recovery execution per FCOM guidance
- TEM principals to optimize crew resources in recognizing and recovering from an approach to stall conditions
- Departure Stall conducted at heavy weight (390K) to replicate a previous FCR

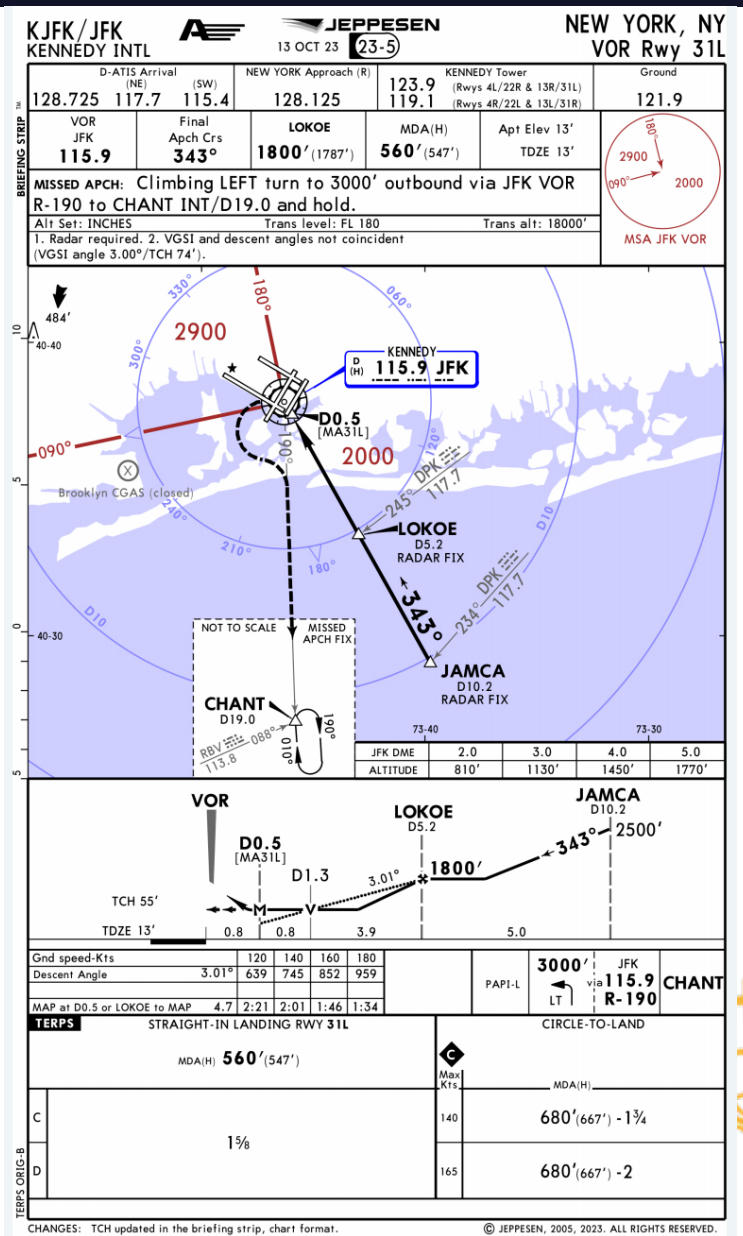
For Training ONLY (FAA Requirement for training)

Stall recovery will be initiated at the first indication of:

- Stick Shaker (Artificial) or,
- Buffet (Natural)

SPOT Review - JFK VOR Rwy 31L

- Runway alignment is a prime concern
- Plan ahead for wind drift
- Do you have to wait until <u>minimums</u> to maneuver for the runway once the runway is in sight?
- Are we still “STABLE” while maneuvering below 1000ft?
- Stabilized Criteria is always a high concern, if in doubt, Go-Around

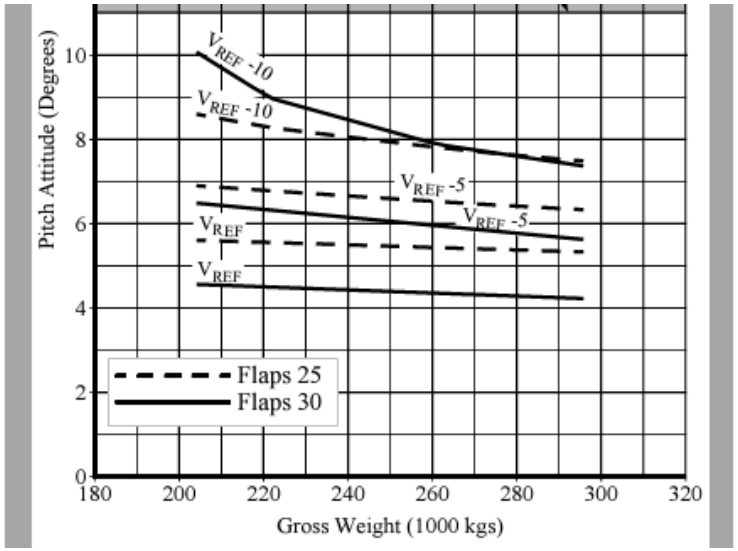


SPOT Review - JFK VOR Rwy 31L

Runway alignment is a prime concern continued…

- Last minute corrections are not advised
- Maneuvering below Vref significantly increases your chance of a pod strike
- If in doubt…

Go Around

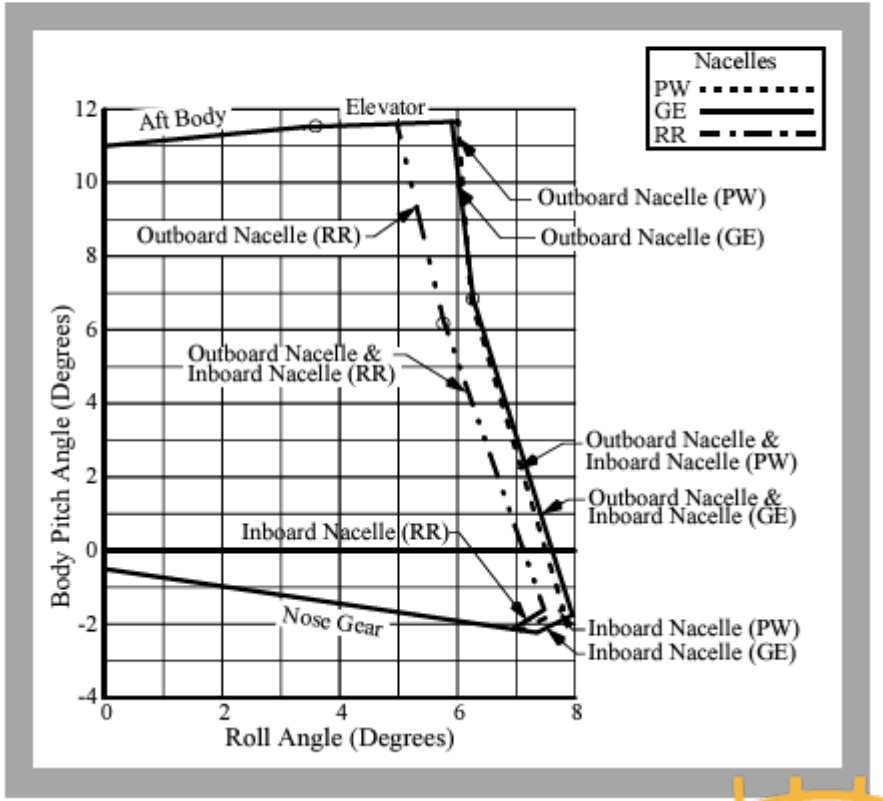


BOEING

747 Flight Crew Training Manual

Ground Contact Angles - Normal Landing

747-400



SPOT Review - Visual Traffic Patern

Review FCOM Vol 1
MAN.1.16 / MAN.2.7

Maneuvers -
Non–Normal Maneuvers

VISUAL APPROACH

General

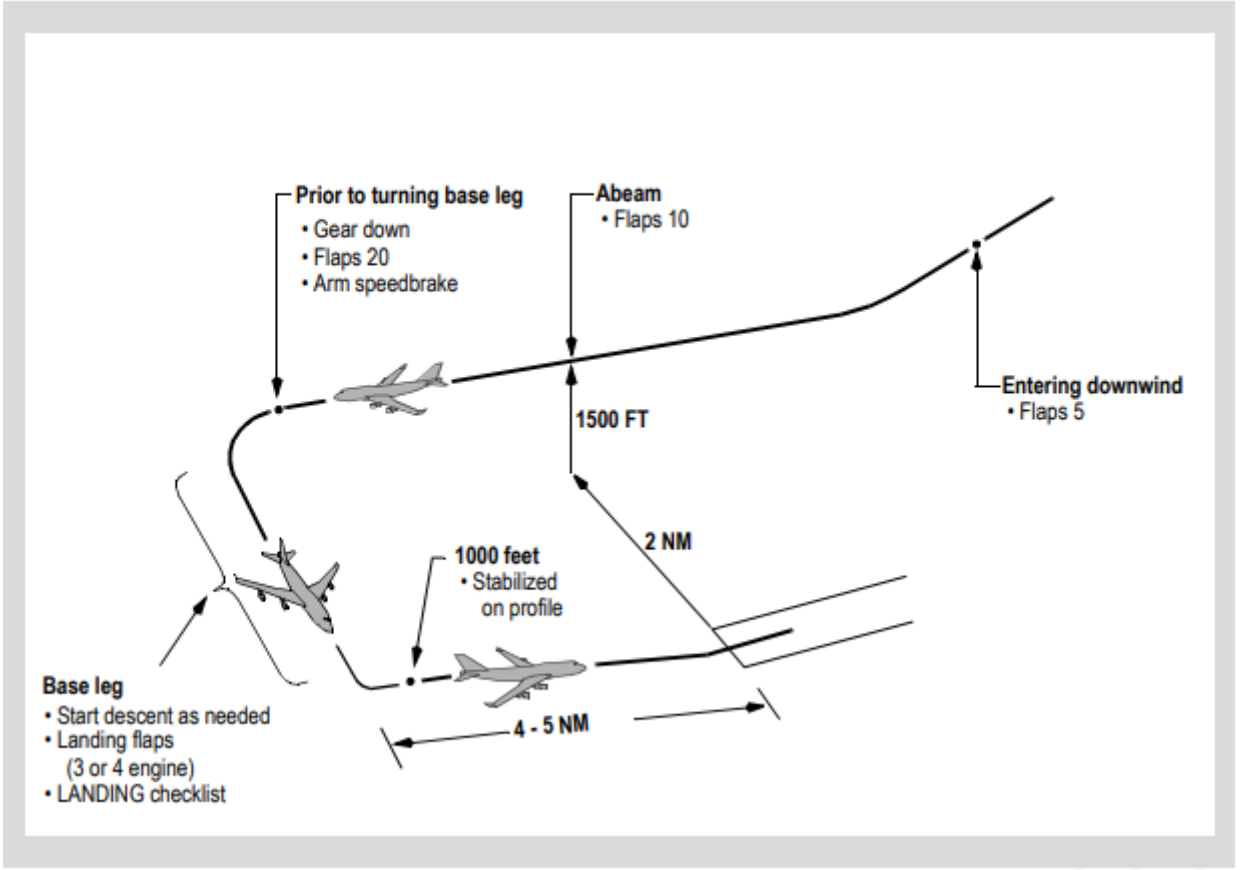
A visual approach is conducted under an IFR flight plan. It allows the pilot to proceed visually to the airport without following a prescribed approach procedure. If the aircraft being followed is in sight, the pilot is responsible for visual separation from other aircraft and wake turbulence avoidance.

Lateral Navigation

The runway is the primary lateral reference on a visual approach. The AFDS can provide important information during approaches with crosswinds or reduced visibility. Displaying the MAP mode with terrain selected can also greatly enhance situational awareness.

Downwind and Base Leg

The following is general guidance from the Boeing Flight Crew Training Manual which assumes the aircraft is remaining in the airport traffic area after takeoff with the intent of entering a downwind for landing. When conducting line operations it will be difficult to adhere to recommended altitudes and turn points described below. Sound judgment should prevail; adjustments must be made which are dependent on altitude and distance from the runway.



SPOT Review - Balked Landing

Boeing FCTM 6.19

Recent FCRs have shown an uptick in balked landings and associated pod/tail strikes resulting from this type of discontinued landing.
Emphasis must be placed on maintaining wind correction, bank, and pitch angles within their respective limits. Allowing an aircraft to drift from the runway centerline could also result in a ground collision.

Rejected Landing / Balked Landing

Rejected Landing

A rejected landing is a discontinued landing attempt and go-around initiated at low altitude (below DA(H) or MDA(H)) but prior to touchdown. If a rejected landing becomes necessary, follow the Go-Around and Missed Approach procedure as prescribed in the FCOM/QRH.

Note: When landing, the TO/GA switches are inhibited near touchdown and on the ground. Refer to the FCOM, Automatic Flight chapter for details.

Balked Landing

A balked landing is a discontinued landing attempt and go-around initiated at or after touchdown, but prior to initiation of reverse thrust. The balked landing technique maintains landing flap configuration to expedite climb away from the runway environment. Considerations for a balked landing include clearance of an obstacle in the runway environment, insufficient runway for continued landing, or transitioning to a go-around from a low-energy state.

Note: When landing, the TO/GA switches are inhibited near touchdown and on the ground. Refer to the FCOM, Automatic Flight chapter for details.

When performing a balked landing, disengage the autopilot, disconnect the autothrottle, smoothly advance thrust levers to go-around thrust, and verify speedbrakes are retracted. Maintain landing flaps configuration and smoothly rotate toward 15° pitch attitude at no less than VREF. Column forces during rotation can vary. When safely airborne with a positive rate of climb, continue the Go-Around and Missed Approach procedure as prescribed in the FCOM/QRH.

WARNING: After reverse thrust is initiated, a full stop landing must be made. If an engine stays in reverse, safe flight is not possible.

Note: The takeoff configuration warning siren sounds when on the ground due to landing flap configuration.

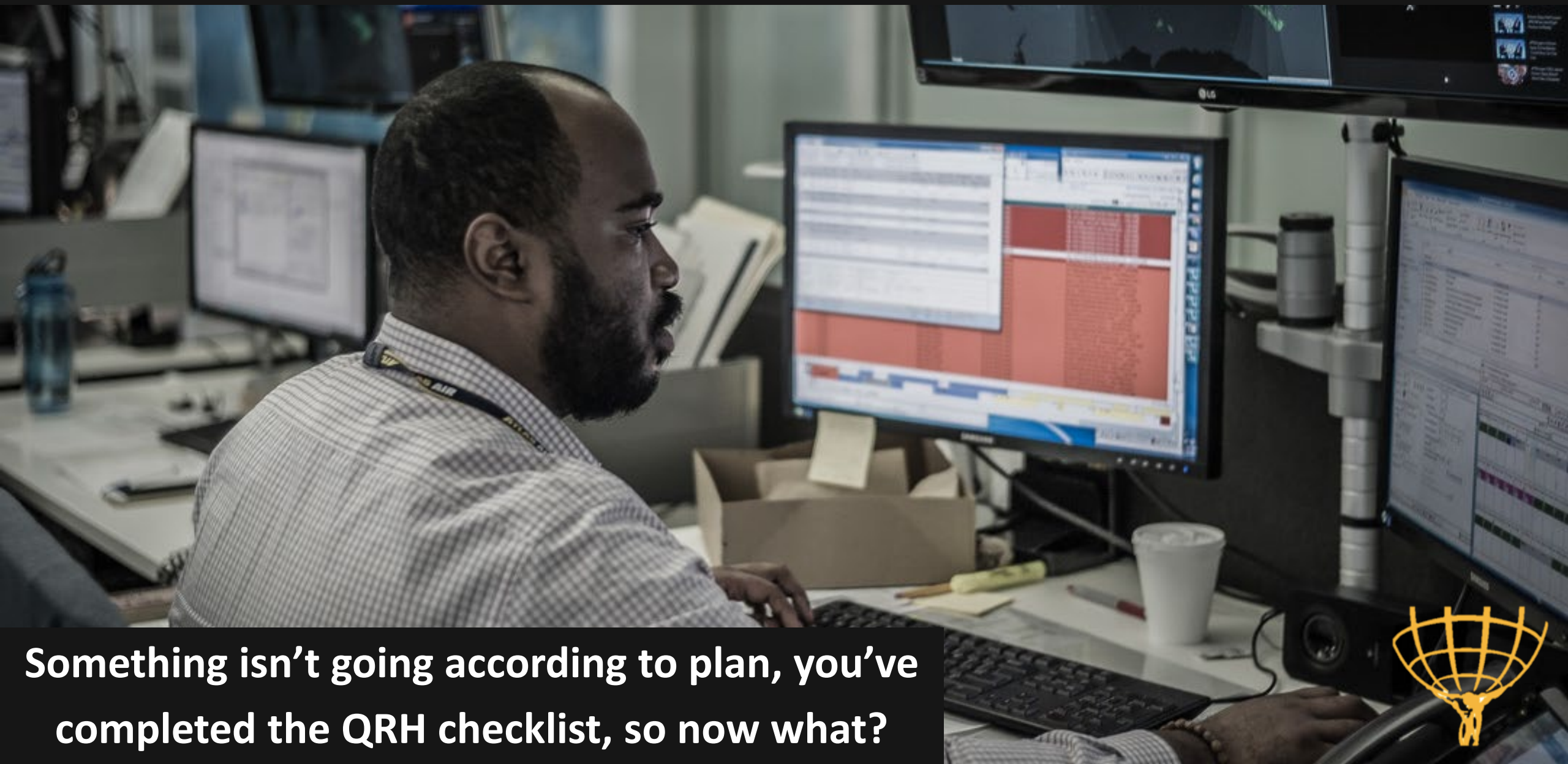
Selecting the go-around flaps during a balked landing can cause a decrease in the airplane lift and can result in increased takeoff distance. However, if there is sufficient runway remaining to safely complete the balked landing with the go-around flap setting, the go-around flap setting can be selected.



BUILDING YOUR TEAM

ATLAS AIR
WORLDWIDE

Who’s on your team?



Something isn’t going according to plan, you’ve completed the QRH checklist, so now what?


How does the Dispatcher play a role once the aircraft is under control?

Expand the CRM Team



Multiple

IN





Multiple

OUT





RLOFT CONCEPTS

ATLAS AIR
WORLDWIDE

Safe Operations



CRM 7 Critical Skills

- Decision Making
- Assertiveness
- Workload Management
- Communication
- Leadership / Teamwork
- Adaptability / Flexibility
- Situational Awareness

TEM is what we do…

…CRM is how we do it

Decision Making/Problem Solving

What are some good life hacks for effective decision making?

- Effective decisions can be made by:
  - Assess
  - Verify
  - Identify
  - Anticipating outcomes
  - Telling others what your thinking
  - Evaluate
- **Improved through** teamwork, extra time, alert crew, decision strategies, and experience



Non-Normal Situation Guidelines Overview

This is a general overview of how non-normal situations should be conducted.

Consider maximum use of autoflight system to reduce workload, if available and appropriate. If flight directors are selected on, ensure the proper flight director modes are selected.

Pilot Flying	Pilot Monitoring
Recognizes and announces the non-normal situation. Other pilot acknowledges.	
MAINTAIN AIRPLANE CONTROL Ensure that the flight path is under control.	Monitor the flight path.
	ANALYZE THE SITUATION Review all EICAS alert messages to identify the non-normal situation. Prioritize alerts and recommend course of action.
Acknowledges and confirms the Pilot Monitoring's identification or recommendation.	
TAKE THE PROPER ACTION Do the NNC immediate action items based on each crewmember's area of responsibility.	
Call for the appropriate NNC.	
	Completes the NNC.
Review all EICAS alert messages, and do other NNCs as needed.	
EVALUATE THE NEED TO LAND Review options for diversion or continued flight.	

"Create the environment for thinking."

Aviate

Navigate

Communicate

Manage



FOR-DEC CHECKLIST

- **FACTS:** What are the facts of the situation?
- **OPTIONS:** What are our options in this situation?
- **RISKS & BENEFITS:** What are the pros and cons?
- **—** Have we gathered the information that we need to continue to a resolution?
- **DECIDE:** What are we going to do? A plan of action.
- **EXECUTE:** Assign roles and carry out the plan.
- **CHECK:** Review the actions taken.

ATLAS AIR

TEM MODEL

UNEXPECTED THREATS

- RECOGNIZE
- STABILIZE
- ORGANIZE
- FOR-DEC

TIME

NO TIME

ATLAS AIR



If we have a NO TIME threat, did we create a contingency plan? (e.g. Engine Failure Procedure or diversion plan during a cruise briefing)

THREATS

EXPECTED | UNEXPECTED | LATENT

01 RECOGNIZE

Situational Awareness

- Determine what is happening
- How will this situation affect the flight?

02 STABILIZE

Flight Path Management (auto or manual)

- Maintain or regain aircraft control

03 ORGANIZE

Workload Management (PF/PM)

- Distribute the workload by clearly assigning PF and PM duties

04 FOR-DEC

Consider time available to resolve:

- Always have a contingency plan for NO TIME threats
- TIME threats do not require immediate crew action, therefore the crew is able to fully evaluate a strategy to mitigate the threat.

TIME

NO TIME

ATLAS AIR

RLOFT Basics

- The RLOFT today will simulate Line Operations
- Must do 2 legs today
- Keep it Real. Address all EICAS and Status messages as you would the real Aircraft.
- Time is built in for required maneuvers completion
  i.e. a go around, holding, and landings, etc. (data is telling us, the more landings we get, the better, we all are.)
- Don’t worry about time, we will manage that. However, please expedite your preflight and engine start as if you are meeting a curfew for departure.

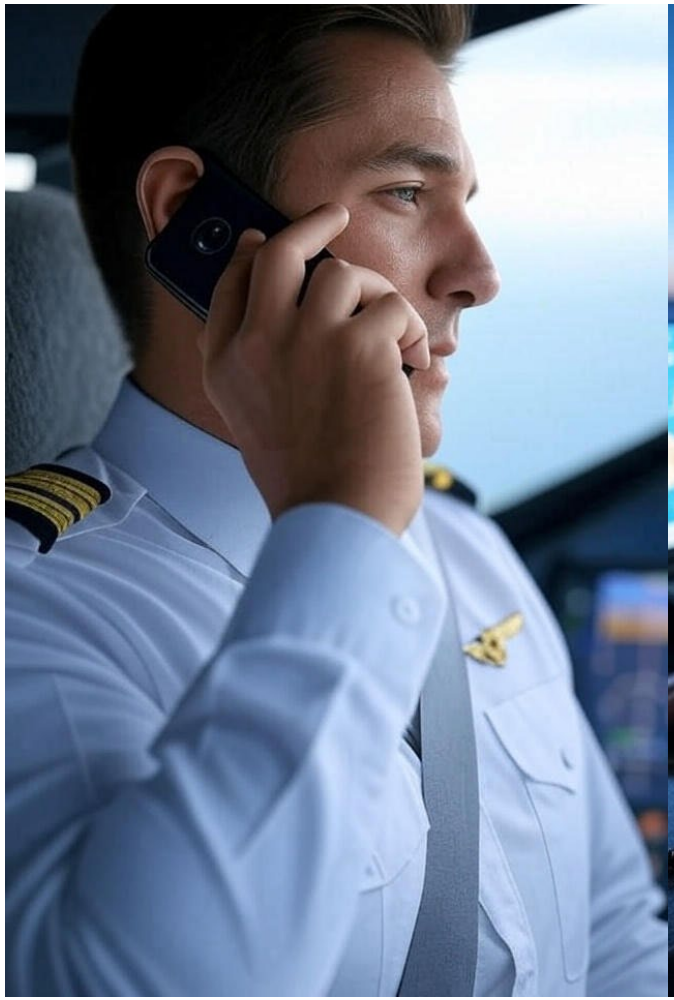


Facilitator (Instructor) Role

- **Once the pre-briefing is complete, I am NOT the instructor anymore.**

I can be:

- **Dispatch**
- **Scheduling**
- **Maintenance**
- **A flight attendant**
- **The PSR, etc.**





Facilitator (Instructor) Role

- To facilitate debriefing as a crew:
  - To capture the mindset of the crew and compare it to your mindset (SOPs) in group discussion.
  - To understand why decisions were made, actions taken and the thought process as a crew.
  - Crew self analysis.

